AI-Generated Text Detection Using Hybrid NLP and Transformer Models

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Abstract-- **Artificial Intelligence generated text has become increasingly difficult to distinguish from human-written content due to the rapid advancement of Large Language Models (LLMs) such as ChatGPT, Gemini, and Claude. This raises major concerns regarding academic integrity, misinformation, and plagiarism detection. Existing AI text detection methods often suffer from poor generalization, high computational complexity, and vulnerability to paraphrasing attacks. This paper presents a hybrid framework for AI-generated text detection using Natural Language Processing (NLP) techniques and transformer-based deep learning models. The proposed system integrates linguistic feature extraction, stylometric analysis, TF-IDF vectorization, and contextual semantic understanding using transformer architectures such as DistilBERT and RoBERTa. The framework is evaluated using benchmark datasets including HC3 and DAIGT v2 under topic-based data splitting to improve robustness and prevent data leakage. Experimental results demonstrate that transformer-based models significantly outperform traditional machine learning approaches in terms of accuracy, recall, and contextual understanding. The proposed hybrid approach improves detection robustness while maintaining scalability and computational efficiency.**

I. INTRODUCTION

The rapid advancement of Artificial Intelligence (AI) and Large Language Models (LLMs) has significantly transformed the way textual content is generated, processed, and consumed across various domains. Modern transformer-based models such as ChatGPT, Gemini, Claude, and other generative AI systems are capable of producing highly coherent and contextually accurate text that closely resembles human-written content. These systems are widely used in academic writing, content generation, summarization, virtual assistance, software development, and research applications. Although these technologies improve productivity and accessibility, they also introduce serious concerns related to misinformation, plagiarism, fake content generation, and academic integrity.The increasing use of AI-generated text in educational and professional environments has created major challenges in distinguishing machine-generated content from authentic human writing. Traditional plagiarism detection systems are often ineffective against modern AI-generated text because such systems primarily rely on direct textual similarity rather than contextual and semantic analysis. Furthermore, rapidly evolving LLM architectures continuously improve the quality of generated content, making detection increasingly difficult. As a result, robust and scalable AI-generated text detection systems have become an important area of research in Natural Language Processing (NLP) and machine learning.Existing AI text detection methods mainly rely on either traditional machine learning techniques or deep learning-based transformer models. Traditional approaches commonly use stylometric and linguistic features such as TF-IDF vectors, readability scores, vocabulary richness, punctuation frequency, and part-of-speech tagging for classification. While these methods provide computational efficiency, they often fail to capture deep contextual relationships within text data. On the other hand, transformer-based architectures such as BERT, DistilBERT, and RoBERTa demonstrate superior contextual understanding and semantic analysis capabilities, resulting in higher detection accuracy and robustness.This paper proposes a hybrid AI-generated text detection framework that integrates traditional NLP-based linguistic feature extraction with transformer-based semantic modeling techniques. The proposed approach combines stylometric analysis, TF-IDF vectorization, and contextual embeddings to improve detection performance across diverse datasets and unseen topics. Benchmark datasets including HC3 and DAIGT v2 are utilized for evaluation under topic-based splitting strategies to reduce information leakage and improve generalization capability. The proposed framework aims to provide an accurate, scalable, and robust solution for identifying AI-generated content in modern digital and academic environments.

A. Background and Motivation

The rapid development of Large Language Models (LLMs) has revolutionized modern Natural Language Processing applications. Advanced generative AI systems such as ChatGPT, Gemini, Claude, and LLaMA are capable of generating highly coherent and contextually meaningful textual content that closely resembles human-written language. These models are widely utilized in academic writing, software development, customer support systems, digital marketing, virtual assistance, and automated content generation platforms.The increasing accessibility of generative AI technologies has significantly improved productivity and automation across multiple industries. However, the misuse of AI-generated content has also created serious challenges related to misinformation, fake news generation, academic dishonesty, plagiarism, and content authenticity verification. AI-generated text can often bypass conventional plagiarism detection systems because modern language models generate unique and contextually rich content rather than directly copying existing textual sources.As the quality of generated content continues to improve, distinguishing AI-generated text from human-written content becomes increasingly difficult. This has created an urgent need for robust and scalable AI-generated text detection systems capable of identifying machine-generated content across diverse domains and writing styles.

B. Challenges in AI-Generated Text Detection

AI-generated text detection presents several technical and practical challenges in modern NLP research. Traditional machine learning techniques mainly rely on statistical and lexical features such as word frequency, sentence structure, punctuation usage, and readability metrics. Although these approaches provide computational efficiency, they often fail to capture deep semantic relationships and contextual understanding within textual data.Modern transformer-based language models continuously improve their ability to mimic human writing patterns, making detection significantly more difficult. AI-generated content can also evade detection systems through paraphrasing, prompt engineering, and adversarial text modification techniques. Additionally, many existing detection systems suffer from poor generalization across unseen topics, datasets, and newly emerging Large Language Models.Another major challenge involves balancing detection accuracy with computational efficiency. Transformer-based architectures such as BERT and RoBERTa achieve high classification performance but require substantial computational resources during training and deployment. Therefore, hybrid frameworks that combine lightweight NLP feature extraction techniques with transformer-based contextual semantic analysis have become an important research direction for improving robustness, scalability, and real-world applicability.

II. LITERATURE REVIEW

A. Traditional Machine Learning Approaches

Early research in AI-generated text detection primarily focused on traditional machine learning techniques using statistical and stylometric features. Researchers utilized features such as TF-IDF vectors, vocabulary richness, readability scores, punctuation frequency, sentence complexity, and part-of-speech tagging for classification tasks. Machine learning models including Logistic Regression, Support Vector Machines (SVM), Random Forest, and XGBoost were widely used due to their computational efficiency and simpler implementation. Studies demonstrated that these approaches achieved moderate detection accuracy; however, they struggled to capture deep contextual and semantic relationships present in modern AI-generated text.

B. Transformer-Based Detection Models

Recent advancements in transformer architectures have significantly improved the performance of AI-generated text detection systems. Models such as BERT, DistilBERT, and RoBERTa utilize self-attention mechanisms and contextual embeddings to understand semantic relationships between words and sentences. Unlike traditional machine learning approaches, transformer-based models can capture contextual meaning and long-range dependencies within textual data. Several studies demonstrated that transformer-based architectures achieved higher accuracy, precision, and recall compared to conventional NLP techniques. RoBERTa-based detection systems achieved near-perfect classification performance on benchmark datasets, highlighting the effectiveness of transformer models in identifying AI-generated content across diverse domains and writing styles.

C. Hybrid NLP Detection Systems

Several researchers proposed hybrid AI-generated text detection frameworks that combine traditional linguistic analysis with deep learning architectures. These systems integrate stylometric analysis, readability metrics, TF-IDF vectorization, semantic embeddings, and transformer-based contextual representations to improve detection robustness and scalability. Hybrid approaches attempt to balance computational efficiency with contextual understanding by combining lightweight NLP techniques and advanced transformer models. Recent studies demonstrated that hybrid frameworks provide improved generalization capability and stronger resistance against paraphrasing attacks and adversarial text modifications. The integration of linguistic features with contextual semantic analysis significantly improves overall detection performance across diverse datasets and unseen topics.

D. Research Gaps

Despite significant advancements in AI-generated text detection, several research challenges still remain unresolved. Many existing systems suffer from poor generalization across different datasets and evolving Large Language Models (LLMs). Traditional machine learning methods often fail to capture contextual semantic relationships, while transformer-based architectures require high computational resources for training and deployment. Additionally, modern AI-generated text can bypass several detection systems through paraphrasing and linguistic manipulation techniques. Most existing studies primarily focus on specific datasets or individual transformer models, limiting scalability and real-world applicability. These limitations highlight the need for a robust hybrid framework capable of integrating linguistic feature extraction with transformer-based semantic analysis for scalable, accurate, and generalized AI-generated text detection.

III. PROPOSED METHODOLOGY

A. System Overview

The proposed system presents a hybrid AI-generated text detection framework that combines traditional Natural Language Processing (NLP) techniques with transformer-based deep learning architectures. The framework is designed to identify whether a given text sample is human-written or AI-generated by analyzing both linguistic patterns and contextual semantic relationships. The proposed approach integrates stylometric feature extraction, TF-IDF vectorization, and transformer-based contextual embeddings to improve detection robustness, scalability, and generalization capability across diverse datasets and unseen topics. The overall workflow of the proposed system consists of multiple stages including data collection, preprocessing, feature extraction, transformer-based semantic analysis, classification, and evaluation. Initially, textual data from benchmark datasets is collected and preprocessed using various NLP techniques such as tokenization, stop-word removal, lemmatization, and part-of-speech tagging. Linguistic and statistical features are then extracted using TF-IDF vectorization and stylometric analysis. Finally, transformer-based models such as DistilBERT and RoBERTa are utilized for contextual semantic understanding and classification of text into human-written or AI-generated categories.

Input Text
↓
Data Preprocessing
↓
Feature Extraction
↓
TF-IDF & Stylometric Analysis
↓
Transformer Models
(DistilBERT / RoBERTa)
↓

Classification
↓
Human Written / AI Generated

Fig. 1. Proposed Hybrid AI-Generated Text Detection Framework

B. Dataset Collection

The proposed framework utilizes benchmark datasets commonly used in AI-generated text detection research, including the HC3 dataset and the DAIGT v2 dataset. The HC3 dataset contains paired human-written and ChatGPT-generated responses collected from multiple domains such as Reddit, finance, medicine, and open-domain question answering. This dataset provides balanced samples of both human and AI-generated content, making it suitable for comparative analysis and supervised classification tasks. The DAIGT v2 dataset contains essays and textual samples generated by multiple Large Language Models (LLMs) alongside human-written content. Unlike HC3, DAIGT v2 includes diverse writing styles, topic variations, and multiple AI-generation sources, improving dataset diversity and generalization capability. The dataset includes topic-specific categories that help evaluate model performance across unseen domains and writing scenarios. To improve robustness and reduce information leakage, topic-based data splitting is utilized instead of random splitting methods. In this approach, entire topics are assigned separately to training, validation, and testing datasets. Topic-based splitting prevents models from memorizing topic-specific vocabulary and improves real-world generalization capability during AI-generated text detection.

C. Data Preprocessing and Feature Extraction

Before model training, textual data undergoes multiple preprocessing stages to improve data quality and feature representation. Initially, all text samples are converted into lowercase format and cleaned by removing special characters, punctuation noise, and unnecessary whitespace. NLP preprocessing techniques such as tokenization, stop-word removal, and lemmatization are then applied to normalize textual data and reduce redundancy. Part-of-speech (POS) tagging is also utilized to identify grammatical patterns and linguistic structures within text samples. After preprocessing, multiple linguistic and statistical features are extracted for classification. TF-IDF vectorization is used to convert textual data into numerical feature representations based on word importance and frequency distribution. Stylometric features such as sentence length, vocabulary richness, readability scores, punctuation frequency, and lexical diversity are also extracted to capture writing patterns commonly associated with AI-generated text. In addition to traditional NLP features, transformer-based contextual embeddings are generated using models such as DistilBERT and RoBERTa. These embeddings capture semantic relationships and contextual meaning within textual data, improving the model's ability to distinguish human-written content from AI-generated text. The integration of linguistic feature extraction and contextual semantic analysis forms the foundation of the proposed hybrid detection framework.

D. Mathematical Formulation

Mathematical and statistical formulations are utilized in the proposed framework to represent textual features and evaluate classification performance. TF-IDF vectorization is used to convert textual data into numerical feature representations based on word importance and frequency distribution across documents. The TF-IDF score for a term is calculated using the following formulation:

$$TF\text{-}IDF(t,d) = TF(t,d) \times \log\left(\frac{N}{DF(t)}\right)$$

where TF(t,d) represents the term frequency of term t in document d, N represents the total number of documents, and DF(t) represents the document frequency of term t across the dataset. TF-IDF helps identify important and discriminative words within textual samples during feature extraction.

The performance of the proposed classification framework is evaluated using standard evaluation metrics such as Accuracy, Precision, Recall, and F1-score. Classification accuracy is computed using the following equation:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP represents True Positives, TN represents True Negatives, FP represents False Positives, and FN represents False Negatives. These evaluation metrics help measure the effectiveness, robustness, and classification capability of the proposed hybrid AI-generated text detection framework.

E. Transformer-Based Classification

Transformer-based architectures are utilized in the proposed framework to perform contextual semantic analysis and text classification. Models such as DistilBERT and RoBERTa are fine-tuned using benchmark datasets to classify textual samples as either human-written or AI-generated. These transformer models utilize self-attention mechanisms that enable contextual understanding of words and sentence relationships within textual data.

DistilBERT is selected due to its lightweight architecture and computational efficiency while maintaining high classification performance. RoBERTa is utilized for its superior contextual representation and strong semantic modeling capability. During training, textual samples are tokenized and converted into transformer-compatible embeddings before being processed through multiple transformer layers for feature learning and classification. The final classification layer predicts whether the input text is AI-generated or human-written based on contextual semantic understanding and extracted linguistic features. The integration of transformer-based classification with NLP feature extraction improves detection robustness, scalability, and overall classification accuracy across diverse datasets and unseen domains.

F. Evaluation Metrics

The performance of the proposed AI-generated text detection framework is evaluated using multiple classification metrics including Accuracy, Precision, Recall, F1-score, and ROC-AUC score. Accuracy measures the overall correctness of classification results, while Precision evaluates the proportion of correctly identified AI-generated samples among all predicted AI-generated texts. Recall measures the ability of the model to correctly identify actual AI-generated content, whereas the F1-score provides a balanced evaluation of Precision and Recall. ROC-AUC is utilized to evaluate the overall classification capability and robustness of the proposed framework across different classification thresholds. These evaluation metrics provide comprehensive performance analysis for both traditional machine learning approaches and transformer-based architectures. The experimental evaluation helps compare classification performance, contextual understanding capability, and generalization efficiency across different AI-generated text detection models.

IV. RESULTS AND DISCUSSION

A. Performance Comparison of Detection Models

The proposed hybrid AI-generated text detection framework was evaluated using both traditional machine learning algorithms and transformer-based architectures. Experimental analysis was conducted using benchmark datasets including HC3 and DAIGT v2 under topic-based splitting conditions to evaluate model robustness and generalization capability. Performance comparison was carried out using multiple evaluation metrics including Accuracy, Precision, Recall, and F1-score.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	82%	81%	80%	80%
BiLSTM	88%	89%	86%	86%
DistilBERT	94%	93%	92%	92%
RoBERTa	98%	97%	97%	97%

TABLE I: PERFORMANCE COMPARISON OF DETECTION MODELS

The experimental analysis also demonstrated that hybrid NLP and transformer-based frameworks improve robustness and generalization capability across unseen topics and diverse writing styles. Topic-based dataset splitting further reduced information leakage and improved real-world evaluation performance. The integration of linguistic feature extraction with transformer-based semantic analysis significantly enhanced AI-generated text detection performance across multiple benchmark datasets.

B. Advantages and Limitations of the Proposed Framework

The proposed hybrid AI-generated text detection framework provides several advantages compared to traditional machine learning approaches and standalone transformer-based architectures. The integration of linguistic feature extraction with transformer-based semantic modeling improves contextual understanding, robustness, and classification accuracy across diverse datasets and writing styles. Traditional NLP techniques such as TF-IDF vectorization and stylometric analysis provide lightweight statistical representations, while transformer models contribute deeper contextual semantic understanding. Another major advantage of the proposed framework is improved generalization capability through topic-based dataset splitting. Unlike random data splitting methods, topic-based evaluation reduces information leakage and improves real-world detection performance across unseen domains. The hybrid framework also demonstrates improved robustness against paraphrasing attacks and adversarial text manipulation techniques commonly used to bypass AI-generated text detection systems. Despite these advantages, several limitations still exist within the proposed framework. Transformer-based architectures such as DistilBERT and RoBERTa require high computational resources during training and fine-tuning. Large-scale deployment may increase memory consumption and inference time in real-time applications. Additionally, rapidly evolving Large Language Models continuously improve text generation quality, making future AI-generated content increasingly difficult to distinguish from human-written text. The performance of AI-generated text detection systems also depends heavily on dataset quality, topic diversity, and model generalization capability. Future improvements should focus on lightweight transformer optimization, multilingual detection, adversarial robustness, and scalable deployment strategies for large-scale educational and digital content verification systems.

C. Comparative Analysis of Detection Models

A comparative analysis was conducted to evaluate the performance, computational efficiency, contextual understanding capability, and robustness of different AI-generated text detection models. Traditional machine learning models, deep learning architectures, and transformer-based frameworks were analyzed based on their detection performance across benchmark datasets.

Model	**Contextual Understanding**	**Computational Efficiency**	**Robustness**	**Accuracy**	**Model**
Logistic Regression	Low	High	Moderate	82%	Logistic Regression
BiLSTM	Moderate	Moderate	Moderate	88%	BiLSTM
DistilBERT	High	Moderate	High	94%	DistilBERT
RoBERTa	Very High	Low	Very High	98%	RoBERTa

TABLE II: COMPARATIVE ANALYSIS OF AI-GENERATED TEXT DETECTION MODELS

The comparative analysis demonstrates that transformer-based architectures significantly outperform traditional machine learning approaches in contextual semantic understanding and AI-generated text classification performance. Logistic Regression provides high computational efficiency but lacks deep contextual analysis capability. BiLSTM improves sequential contextual learning but still faces limitations in capturing long-range semantic dependencies. DistilBERT achieves a strong balance between computational efficiency and classification performance due to its lightweight transformer architecture. RoBERTa demonstrates superior contextual representation and robustness but requires higher computational resources during training and deployment. The proposed hybrid framework combines the advantages of traditional NLP techniques and transformer-based semantic modeling, resulting in improved robustness, scalability, and overall detection performance across diverse datasets and unseen domains.

V. FUTURE RESEARCH DIRECTIONS

The rapid advancement of Large Language Models (LLMs) continuously introduces new challenges for AI-generated text detection systems. Future research should focus on developing more generalized and adaptive hybrid detection frameworks capable of identifying content generated by emerging transformer architectures and multimodal AI systems. As generative AI models become increasingly sophisticated, detection systems must evolve to improve contextual understanding, semantic analysis, and adversarial robustness. One important future research direction involves multilingual AI-generated text detection. Most existing detection systems are primarily trained on English-language datasets, limiting their applicability across multilingual and cross-cultural digital environments. Developing multilingual transformer-based detection systems capable of analyzing diverse linguistic structures and writing styles will improve scalability and global usability. Another promising research area involves lightweight transformer optimization for real-time deployment. Although transformer-based architectures provide high detection accuracy, they often require substantial computational resources during training and inference. Future studies can explore model compression, knowledge distillation, and efficient transformer architectures to improve scalability and reduce computational complexity.

Future AI-generated text detection systems may also integrate explainable artificial intelligence (XAI) techniques to improve model interpretability and transparency. Explainable detection frameworks can help users understand the reasoning behind classification decisions, improving trust and usability in educational, academic, and digital content verification systems.

V. CONCLUSION AND FUTURE WORK

This paper presented a hybrid AI-generated text detection framework that combines traditional Natural Language Processing (NLP) techniques with transformer-based deep learning architectures for robust and scalable AI-generated text classification. The proposed system integrates stylometric analysis, TF-IDF vectorization, and transformer-based semantic embeddings to improve contextual understanding and detection accuracy across diverse datasets and unseen topics. Experimental analysis demonstrated that transformer-based architectures such as DistilBERT and RoBERTa significantly outperform traditional machine learning approaches in terms of contextual semantic understanding, classification accuracy, and generalization capability. The integration of linguistic feature extraction with transformer-based semantic modeling improved robustness against paraphrasing attacks and linguistic modifications in AI-generated content. The proposed hybrid framework provides an effective solution for AI-generated text detection in educational, academic, and digital content verification systems. Future research can focus on multilingual AI-generated text detection, lightweight transformer optimization, adversarial robustness, and real-time deployment for large-scale content moderation systems.

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